

// Here is the code

```
import React, { useState, memo } from 'react';
import {
  StyleSheet,
  Text,
  View,
  TouchableOpacity,
  SafeAreaView,
} from 'react-native';

// Recursive Component
const TreeNode = ({
  item, depth = 0, selected, setSelected
}) => {
  const [open, setOpen] = useState(false);

  const hasChild = item?.child?.length;
  console.log('hasChild ----', hasChild);

  const handlePress = () => {
    if (hasChild) {
      setOpen((prev) => !prev);
    }
  };

  return (
    <View>
      <TouchableOpacity
        onPress={handlePress}
        activeOpacity={0.6}
        style={[styles.row, { paddingLeft: depth * 14 }, selected === item.name &&
styles.selectedRow]}
      >
        <Text style={styles.icon}>
          {hasChild ? (open ? '▼' : '►') : '•'}
        </Text>

        <Text style={styles.text}>{item.name}</Text>
      </TouchableOpacity>

      {open && hasChild
        && item.child.map((child, i) => (
          <TreeNode
            key={` ${item.name}-${i}`}
            item={child}
            depth={depth + 1}
          />
        ))
      }
    </View>
  );
};
```

```

    )))}
  </View>
);
};

```

// Tree Wrapper

```

const TreeView = ({ data = [] }) => {
  return (
    <View>
      {data.map((item, index) => (
        <TreeNode
          key={` ${item.name}-${index}`}
          item={item}
        />
      ))}
    </View>
  );
};

```

// Mock Data

```

const treeData = [
  {
    name: 'Applications',
    child: [{ name: 'Calendar', child: [] }],
  },
  {
    name: 'Documents',
    child: [
      { name: 'OSS', child: [] },
      {
        name: 'MUI',
        child: [{ name: 'index.js', child: [] }],
      },
    ],
  },
];

```

// Main Screen

```

const App = () => (
  <SafeAreaView style={styles.container}>
    <Text style={styles.header}>Screening - Assignment</Text>
    <View style={styles.top}>
      <TreeView data={treeData} />
    </View>
  </SafeAreaView>
);

```

```
export default memo(App);

// Styles
const styles = StyleSheet.create({
  container: {
    flex: 1,
    padding: 16,
    alignItems: 'center',
  },
  header: {
    fontSize: 18,
    fontWeight: '600',
    textDecorationLine: 'underline',
    marginBottom: 12,
  },
  row: {
    flexDirection: 'row',
    alignItems: 'center',
    paddingVertical: 6,
  },
  icon: {
    width: 18,
    color: '#444',
  },
  text: {
    fontSize: 15,
  },
  top: {
    marginTop: 10
  },
  selectedRow: {
    backgroundColor: '#d0ebff',
  }
});
```